Physics

1. The fundamental and the derived physical quantities:

1.1. The fundamental physical quantities (F.P.Q)

F.P.Q.	M.U.	Dimension
1. Length	m	L
2. Time	s	T
3. Mass	kg	M
4. Intensity of electric cur.	A	I
5. Intensity of light	cd	J
6. Temperature	K	Θ
7. The plane angle	rad	–
8. The solid angle	s.r.	

1.2. The derived physical quantities:

A. $Q = N[Q]_{SI}$

- $[Q]_{SI}$ — the measurement unit
- N — numeric value
- Q — quantity

$[Q] = L^{\alpha} \cdot T^{\beta} \cdot M^{\delta} \cdot I^{\xi} \cdot J^{\varepsilon} \cdot \Theta^{\nabla}$

$[Q]$ — dimension

We can add or subtract only quantities of the same type

B. Examples:

- $v = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$

$$[v]_{SI} = \left[\frac{\Delta x}{\Delta t}\right]_{SI} = \frac{[\Delta x]_{SI}}{[\Delta t]_{SI}} = \frac{[x]_{SI}}{[t]_{SI}} = \frac{m}{s}$$

v = velocity

$$[v]_{SI} = \frac{m}{s}$$

$$[v] = \left[\frac{\Delta x}{\Delta t}\right] = \frac{[x]}{[t]} = \frac{L}{T}$$

$$[v] = L \cdot T^{-1} = L^1 \cdot T^{-1} \cdot M^0 \cdot I^0 \cdot J^0 \cdot \Theta^0$$

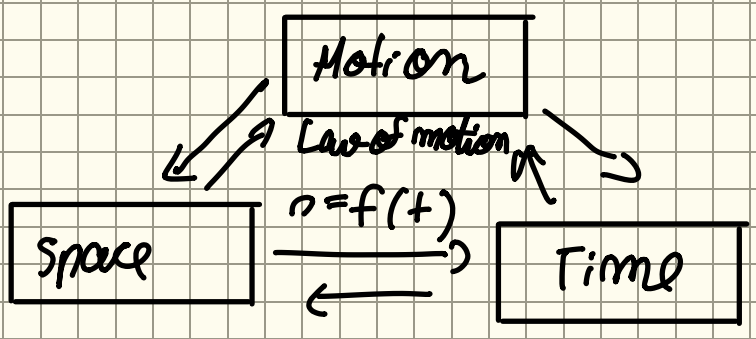
- $a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$

$$[a]_{SI} = \frac{[v]_{SI}}{[t]_{SI}} = \frac{m/s}{s} \Rightarrow [a]_{SI} = m/s^2$$

$$[a] = \frac{[v]}{[t]} = L \cdot T^{-2}$$

$$[a] = L^1 \cdot T^{-2} \cdot M^0 \cdot I^0 \cdot J^0 \cdot \Theta^0$$

2. The fundamental (key) element in physics:



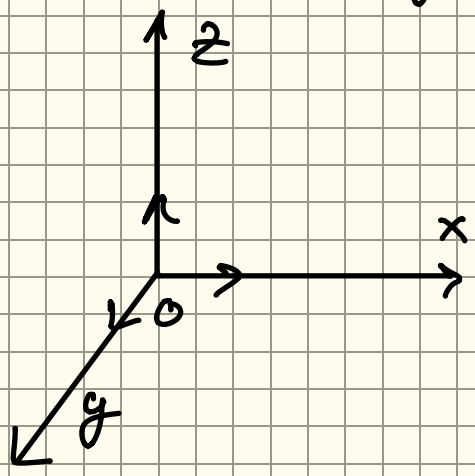
3. Mechanics

- Kinematics → part of the mechanics that studies the motion without interest on causes and effects
- Dynamics → // motion with interest on their nature, masses, causes and effects
- Statics → part of mechanics that study the equilibrium

I Kinematics

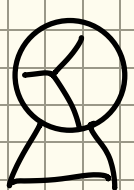
1. The elements of motion

1.1. The reference system (frame)



cartesian system
$|\vec{i}| = |\vec{j}| = |\vec{z}| = 1$

clock



1.2. The trajectory → a curve

